Infinitely many primes congruent to -1 modulo 6
Ireland–Rosen, 3.1

Theorem 1. *There are infinitely many primes congruent to -1 modulo 6.*

Proof. A prime congruent to -1 modulo 6 is the same as a prime p with $p \equiv 5 \pmod 6$. We prove the following stronger statement: for every natural number n, there exists a prime p such that $n < p$ and $p \equiv 5 \pmod 6$. This clearly implies that there are infinitely many such primes.

We first prove an auxiliary claim.

Claim: For every k, and for every $N \leq k$ with $N \equiv 5 \pmod 6$, there exists a prime p such that $p \mid N$ and $p \equiv 5 \pmod 6$.

We prove the claim by induction on k.

If $k = 0$, then $N \leq 0$, so $N = 0$. But $0 \not\equiv 5 \pmod 6$, contradicting the assumption on N. Thus the claim holds vacuously.

Now assume the claim for k, and let $N \leq k+1$ with $N \equiv 5 \pmod 6$. Since $1 \not\equiv 5 \pmod 6$, we have $N \neq 1$. Hence N has a prime divisor p.

We have $p \neq 2$: indeed, if $p = 2$, then $2 \mid N$, so N would be even, whereas $N \equiv 5 \pmod 6$ implies that N is odd. Similarly $p \neq 3$: if $p = 3$, then $3 \mid N$, whereas $N \equiv 5 \pmod 6$ implies $N \equiv 2 \pmod 3$, so $3 \nmid N$.

Since p is prime, neither 2 nor 3 divides p. Indeed, if $2 \mid p$, then primality of p would force $2 = 1$ or $2 = p$, both impossible; likewise, if $3 \mid p$, then $3 = 1$ or $3 = p$, both impossible.

Now consider p modulo 6. Its remainder upon division by 6 is one of $0, 1, 2, 3, 4, 5$. The remainders 0, 2, and 4 would imply that $2 \mid p$, and the remainder 3 would imply that $3 \mid p$. Since both are impossible, we must have

$$p \equiv 1 \pmod 6 \qquad \text{or} \qquad p \equiv 5 \pmod 6.$$

If $p \equiv 5 \pmod 6$, then p itself is the desired prime divisor of N.

Otherwise, suppose $p \equiv 1 \pmod 6$. Write $N = pm$. Modulo 6 we get

$$5 \equiv N = pm \equiv m \pmod 6,$$

since $p \equiv 1 \pmod 6$. Thus $m \equiv 5 \pmod 6$.

Also $m > 0$: if $m = 0$, then $N = pm = 0$, contradicting $N \equiv 5 \pmod 6$. Moreover, since p is prime, $p \geq 2$, and hence

$$m < pm = N,$$

because $m > 0$. Therefore $m < N \leq k+1$, so $m \leq k$. By the induction hypothesis applied to m, there exists a prime q such that $q \mid m$ and $q \equiv 5 \pmod 6$. Since $q \mid m$ and $m \mid N$ (because $N = pm$), we have $q \mid N$. Thus q is the desired prime divisor of N.

This completes the induction and proves the claim.

In particular, taking $k = N$ in the claim, we obtain:

Key fact: Every N with $N \equiv 5 \pmod 6$ has a prime divisor p with $p \equiv 5 \pmod 6$.

We now prove the theorem. Let n be given, and put $F = (n+1)!$. Since $F > 0$, we have

$$6F - 1 = 6(F-1) + 5,$$

so $6F - 1 \equiv 5 \pmod 6$. By the key fact, there exists a prime p such that

$$p \mid 6(n+1)! - 1 \qquad \text{and} \qquad p \equiv 5 \pmod 6.$$

It remains to show that $n < p$. Suppose instead that $p \leq n$. Since p is prime, $p > 0$, and since $p \leq n \leq n+1$, the number p occurs as a factor in $(n+1)!$; hence

$$p \mid (n+1)!.$$

Consequently $p \mid 6(n+1)!$. But we also know that $p \mid 6(n+1)! - 1$. Therefore p divides the difference

$$6(n+1)! - \big(6(n+1)! - 1\big) = 1,$$

that is, $p \mid 1$. Hence $p \leq 1$. This contradicts the fact that the prime p satisfies $p \geq 2$.

Thus $n < p$. So for every n there is a prime $p > n$ with $p \equiv 5 \pmod 6$, i.e. $p \equiv -1 \pmod 6$. Therefore there are infinitely many primes congruent to -1 modulo 6. $\square$